

EDEXCEL INTERNATIONAL A LEVEL

WMA12/01 January 2024 Worked Solutions

January 2024 paper rebuilt with the worked solution after each question.

10

paper questions

WMA12

Pure 2

**Single-
paper
solutions**standalone IAL
route**Dr Eslam Ahmed**Prepared for Dr Eslam Ahmed - eliteigcse.com**P2**

PAST PAPER

WMA12/01 January 2024

January 2024 | 10 questions | 75 marks

10

questions

75

marks

Answers

worked solution
after each
question

Question 1

Polynomials

1. $f(x) = ax^3 + 3x^2 - 8x + 2$ where a is a constant

Given that when $f(x)$ is divided by $(x - 2)$ the remainder is 3, find the value of a .

(3)

(Total for Question 1 is 3 marks)

(Total for Question 1 is 3 marks)

Worked Solution - Question 1

1. Use the polynomial condition

Apply the factor theorem, remainder theorem, or substitution stated in the question to form the required equation.

2. Finish the algebra

Solve the resulting equation and substitute back where needed, then write the requested factor, coefficient, or value.

3. State the final result

Write the answer in the form requested by the question: The required value or expression is obtained by the final simplification.

Final answer

The required value or expression is obtained by the final simplification.

Question 2

Binomial Expansion

2. Find the coefficient of the term in x^7 of the binomial expansion of

$$\left(\frac{3}{8} + 4x\right)^{12}$$

giving your answer in simplest form.

(3)

(Total for Question 2 is 3 marks)

(Total for Question 2 is 3 marks)

Worked Solution - Question 2

1. Choose the required binomial terms

Write the expansion term-by-term and keep only the powers needed for the coefficient or approximation.

2. Compare coefficients

Collect like powers, compare with the given expression, and solve for the required constant.

3. State the final result

Write the answer in the form requested by the question: The required value or expression is obtained by the final simplification.

Final answer

The required value or expression is obtained by the final simplification.

Question 3

Circles

3. The circle C

- has centre $A(3, 5)$
- passes through the point $B(8, -7)$

(a) Find an equation for C .

(3)

The points M and N lie on C such that MN is a chord of C .

Given that MN

- lies above the x -axis
- is parallel to the x -axis
- has length $4\sqrt{22}$

(b) find an equation for the line passing through points M and N .

(3)

(Total for Question 3 is 6 marks)

(Total for Question 3 is 6 marks)

Worked Solution - Question 3

Topic group

1. Identify the circle information

Use the centre-radius form, distance formula, gradient condition, or tangent property required by the diagram.

2. Complete the geometry

Substitute the coordinates carefully and simplify to the requested equation, length, point, or proof.

3. State the final result

Write the answer in the form requested by the question: - = States or

Final answer

- = States or

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9 marks

Question 4

Integration

4. (a) Sketch the curve with equation

$$y = a^{-x} + 4$$

where a is a constant and $a > 1$

On your sketch show

- the coordinates of the point of intersection of the curve with the y -axis
- the equation of the asymptote to the curve.

(3)

x	-4	-1.5	1	3.5	6	8.5
y	13	6.280	4.577	4.146	4.037	4.009

The table above shows corresponding values of x and y for $y = 3^{-\frac{1}{2}x} + 4$

The values of y are given to four significant figures, as appropriate.

Using the trapezium rule with all the values of y in the table,

(b) find an approximate value for

$$\int_{-4}^{8.5} \left(3^{-\frac{1}{2}x} + 4 \right) dx$$

giving your answer to two significant figures.

(3)

(c) Using the answer to part (b), find an approximate value for

(i) $\int_{-4}^{8.5} \left(3^{-\frac{1}{2}x} \right) dx$

(ii) $\int_{-4}^{8.5} \left(3^{-\frac{1}{2}x} + 4 \right) dx + \int_{-8.5}^4 \left(3^{\frac{1}{2}x} + 4 \right) dx$

(3)

(Total for Question 4 is 9 marks)

(Total for Question 4 is 9 marks)

Worked Solution - Question 4

1. Set up the integral

Choose the correct integrand and limits, or apply the trapezium rule if the question gives tabulated values.

2. Evaluate the area or expression

Integrate or calculate the trapezium estimate carefully, then simplify the requested value.

3. State the final result

Write the answer in the form requested by the question: The required value or expression is obtained by the final simplification.

Final answer

The required value or expression is obtained by the final simplification.

Question 5

Sequences & Series

5. (i) Find the value of

$$\sum_{r=1}^{\infty} 6 \times (0.25)^r$$

(3)

- (ii) A sequence $u_1, u_2, u_3, \dots$ is defined by

$$u_1 = 3$$

$$u_{n+1} = \frac{u_n - 3}{u_n - 2} \quad n \in \mathbb{N}$$

- (a) Show that this sequence is periodic.

(2)

- (b) State the order of this sequence.

(1)

- (c) Hence find

$$\sum_{n=1}^{70} u_n$$

(2)

(Total for Question 5 is 8 marks)

(Total for Question 5 is 8 marks)

Worked Solution - Question 5

Topic group

1. Translate the sequence rule

Write the recurrence, sigma expression, or general term exactly from the question before simplifying.

2. Find the requested term or sum

Evaluate the sequence or series in order, then state the required term, total, or conclusion.

3. State the final result

Write the answer in the form requested by the question: $= \times + + + =$

Final answer

$= \times + + + =$

Question 6

Laws of Logarithms

6. (a) Given that

$$2\log_4(x+3) + \log_4 x = \log_4(4x+2) + \frac{1}{2}$$

show that

$$x^3 + 6x^2 + x - 4 = 0 \quad (4)$$

(b) Given also that -1 is a root of the equation

$$x^3 + 6x^2 + x - 4 = 0$$

(i) use algebra to find the other two roots of the equation.

(3)

(ii) Hence solve

$$2\log_4(x+3) + \log_4 x = \log_4(4x+2) + \frac{1}{2} \quad (1)$$

(Total for Question 6 is 8 marks)

(Total for Question 6 is 8 marks)

Worked Solution - Question 6

Topic group

1. Combine the logarithms

Use the power, product, and quotient laws to rewrite the equation as a single logarithmic or exponential statement.

2. Solve with domain checks

Solve the resulting equation and reject any value that does not satisfy the original logarithms.

3. State the final result

Write the answer in the form requested by the question: $x = 5 \cdot 41 \cdot 2^x - 1$

Final answer

$$x = 5 \cdot 41 \cdot 2^x - 1$$

Question 7

Modelling with Sequences & Series

7. Wheat is grown on a farm.

- In year 1, the farm produced 300 tonnes of wheat.
- In year 12, the farm is predicted to produce 4 000 tonnes of wheat.

Model A assumes that the amount of wheat produced on the farm will increase by the same amount each year.

- (a) Using model A , find the amount of wheat produced on the farm in year 4.
Give your answer to the nearest 10 tonnes.

(3)

Model B assumes that the amount of wheat produced on the farm will increase by the same percentage each year.

- (b) Using model B , find the amount of wheat produced on the farm in year 2.
Give your answer to the nearest 10 tonnes.

(3)

- (c) Calculate, according to the two models, the difference between the total amounts of wheat predicted to be produced on the farm from year 1 to year 12 inclusive.
Give your answer to the nearest 10 tonnes.

(3)

(Total for Question 7 is 9 marks)

(Total for Question 7 is 9 marks)

Worked Solution - Question 7

1. Model the situation

Match the real situation to an arithmetic, geometric, or recurrence model using the starting value and rate of change.

2. Interpret the result

Calculate the required term or sum and state the answer in the units and context of the question.

3. State the final result

Write the answer in the form requested by the question: - - Difference = 7860 but

Final answer

- - Difference = 7860 but

Question 8

Proof

8. (i) Use a counter example to show that the following statement is **false**

“ $n^2 + 3n + 1$ is prime for all $n \in \mathbb{N}$ ”

(2)

- (ii) Use algebra to prove by exhaustion that for all $n \in \mathbb{N}$

“ $n^2 - 2$ is **not** a multiple of 4”

(4)

(Total for Question 8 is 6 marks)

(Total for Question 8 is 6 marks)

Worked Solution - Question 8

Topic group

1. Set up the proof

Translate the statement into algebra, or choose a clear counterexample if the statement is false. Keep each implication justified.

2. Complete the argument

Simplify the algebra or evaluate the counterexample, then finish with a clear conclusion.

3. State the final result

Write the answer in the form requested by the question: The statement is false; a valid counterexample completes the disproof.

Final answer

The statement is false; a valid counterexample completes the disproof.

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14 marks

Question 9

Trigonometric Equations

9.

In this question you must show detailed reasoning.

Solutions relying entirely on calculator technology are not acceptable.

- (i) Solve, for $0 \leq x < 360^\circ$, the equation

$$\sin x \tan x = 5$$

giving your answers to one decimal place.

(6)

- (ii)

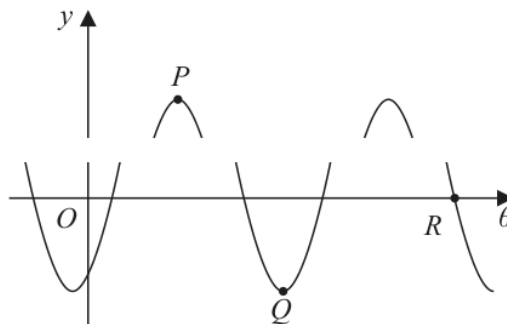


Figure 1

Figure 1 shows a sketch of part of the curve with equation

$$y = A \sin\left(2\theta - \frac{3\pi}{8}\right) + 2$$

where A is a constant and θ is measured in radians.

The points P , Q and R lie on the curve and are shown in Figure 1.

Given that the y coordinate of P is 7

- (a) state the value of A ,

(1)

- (b) find the exact coordinates of Q ,

(3)

- (c) find the value of θ at R , giving your answer to 3 significant figures.

(4)

(Total for Question 9 is 14 marks)

(Total for Question 9 is 14 marks)

Worked Solution - Question 9

1. Rearrange the trigonometric equation

Use the identity or ratio needed to reduce the equation to one trigonometric function in the required interval.

2. List the valid solutions

Find all angles in the given range and discard any extra values outside the interval.

3. State the final result

Write the answer in the form requested by the question: The required value or expression is obtained by the final simplification.

Final answer

The required value or expression is obtained by the final simplification.

Question 10

Applications of Differentiation

10.

In this question you must show detailed reasoning.

Solutions relying entirely on calculator technology are not acceptable.

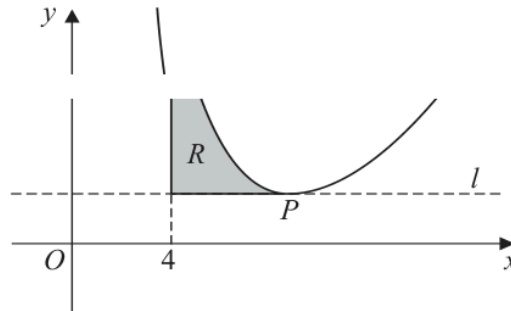


Figure 2

Figure 2 shows a sketch of the curve with equation

$$y = \frac{1}{2}x^2 + \frac{1458}{\sqrt{x^3}} - 74 \quad x > 0$$

The point P is the only stationary point on the curve.(a) Use calculus to show that the x coordinate of P is 9

(4)

The line l passes through the point P and is parallel to the x -axis.The region R , shown shaded in Figure 2, is bounded by the curve, the line l and the line with equation $x = 4$ (b) Use algebraic integration to find the exact area of R .

(5)

(Total for Question 10 is 9 marks)

(Total for Question 10 is 9 marks)

Worked Solution - Question 10

Topic group

1. Differentiate and set the condition

Differentiate the expression, then use the gradient, stationary-point, tangent, normal, or optimisation condition.

2. Classify or evaluate

Substitute the required value and use the derivative information to complete the coordinate, gradient, interval, or maximum/minimum.

3. State the final result

Write the answer in the form requested by the question: \times - \times - \times - \times - \times - \times - - \times 1 1 1 1516 1743 102 2 3 2 -

Final answer

$$\begin{array}{ccccccc} \times & - & \times & - & \times & - & \times \\ 102 & 2 & 3 & 2 & - & & \end{array}$$